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SoftHier Meeting September 29



SoftHier Simulator Workflow

Step For the E2E Verification Pipeline	Implemented?
• Generate SoftHier Architecture Configuration	Yes
• Generate Layout for HBM Arrays, ELF binary to move data to the HBM-Files	Yes
• Run Both SDFG and Reference	Yes
• Copy back data from HBM-files to Host and Compare	Yes

Aofeng is working on re-running the GEMM benchmarks with verification enabled (and fixing bugs)

SoftHier Simulator Workflow

E2E Pipeline Looks like follows

```python
def run_e2e_verification(hw_config: HardwareConfig,
                         data: Dict[str, Any],
                         interleave_handlers: Dict[str, Any],
                         numpy_fn: Callable,
                         sdfg_fn: Callable,
                         tolerance: float = 1e-3) -> Dict[str, Any]:
    # Step 1 Setup
    setup_hw_env_dace(hw_config)

    # Step 2 Copy Data
    numpy_data = data
    sdfg_data = copy.deepcopy(data)

    # Step 3 Run Numpy reference
    numpy_fn()

    # Step 4 Run SoftHier simulator
    sdfg_fn()

    # Step 5 Compare Data
    comparison = compare(hw_config, numpy_data, sdfg_data, interleave_handlers, tolerance)

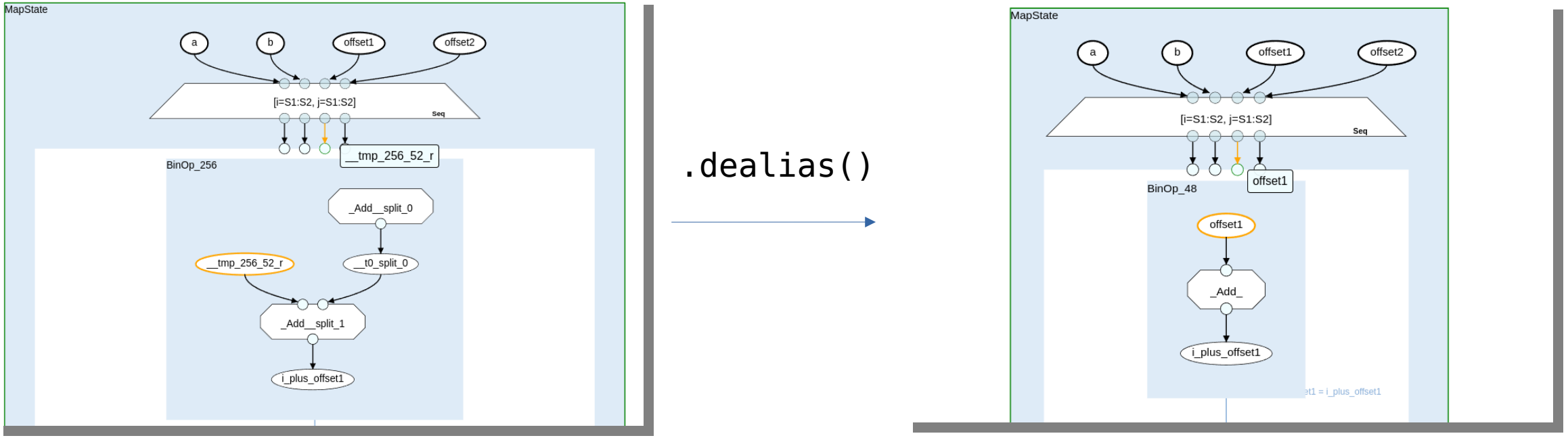
    return comparison['all_match']
```

Implemented Multiple Components of the Explicit Vectorization Pipeline

- Explicit vectorization is partially completed. Multiple components of the pipeline have been implemented.

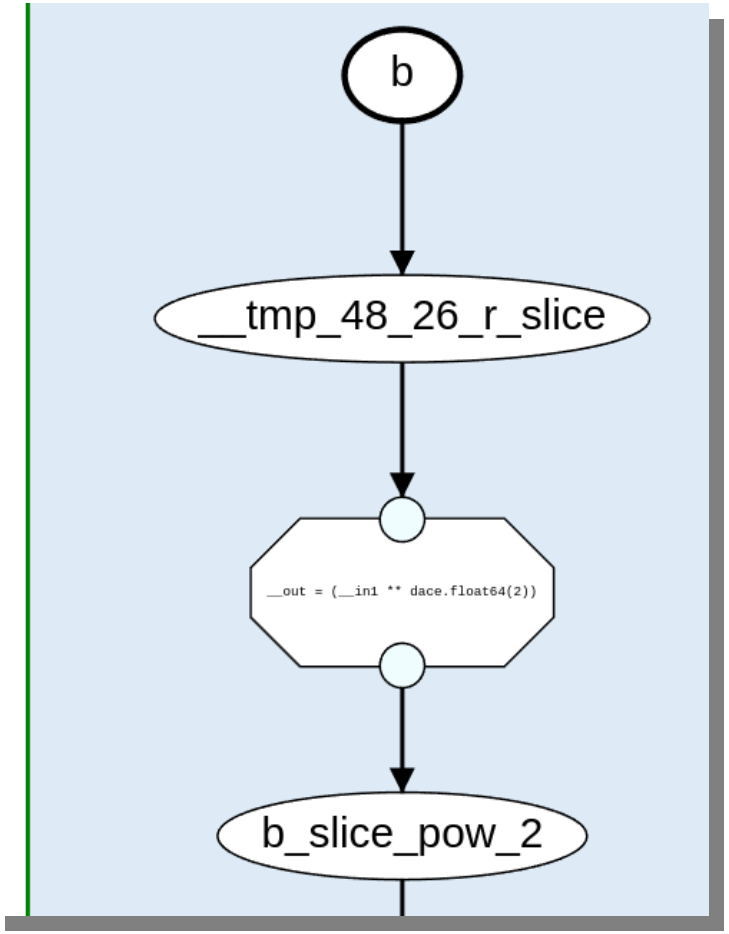
NestedSDFG Dealiasing

- Dealiasing nestedSDFG connectors simplifies bookkeeping.



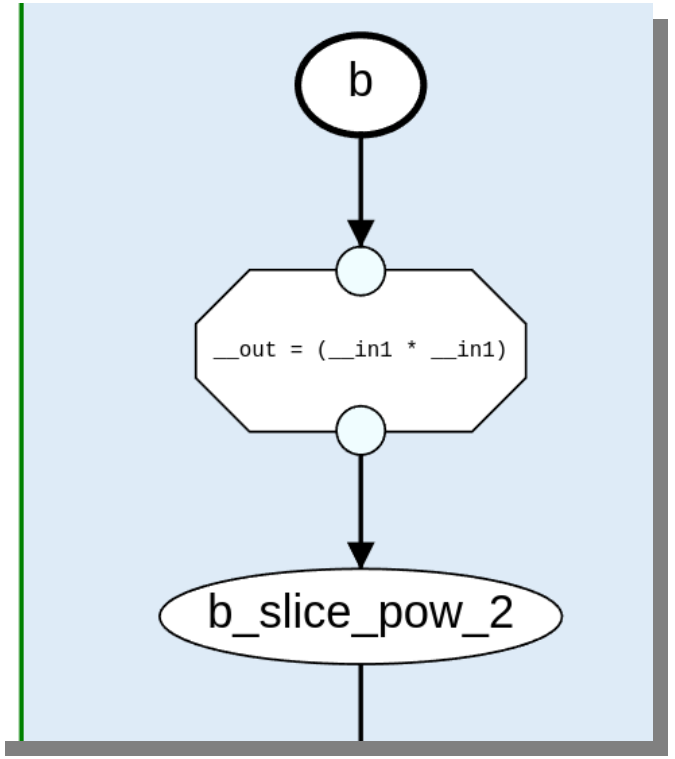
Tasklet Preprocessing / Pruning

- Remove typecasts from tasklets, try to prune unsupported expressions such as `pow()` from tasklets
- Makes SDFGs amenable to vectorization pass



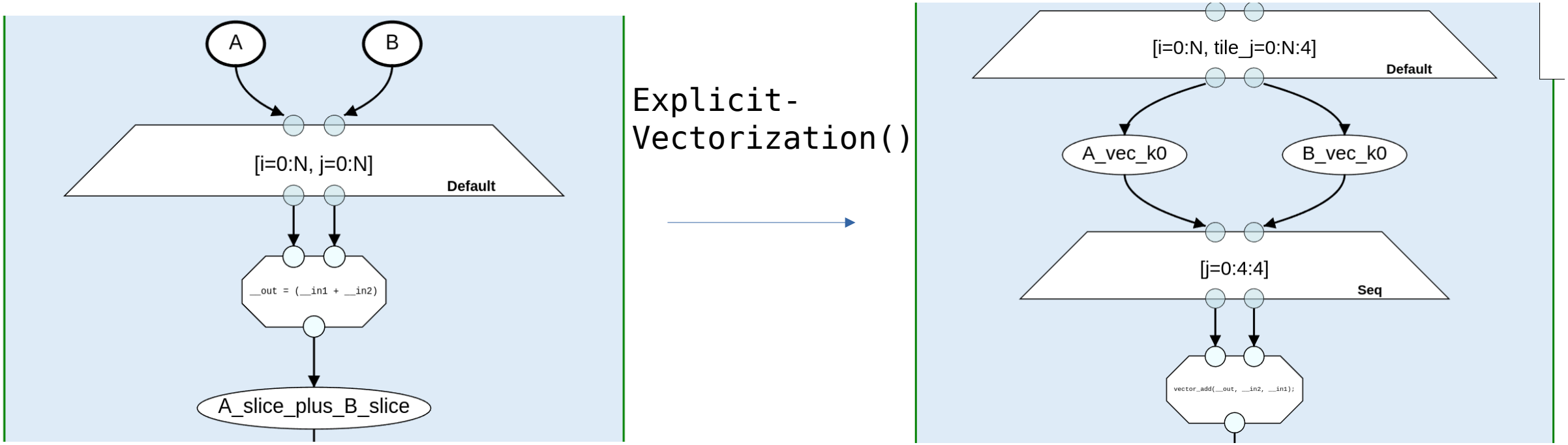

```
RemoveFPTypeCasts()
RemoveIntTypeCasts()
IntegerPowerToMult()

Simplify-
TaskletAssignmentChains()
```



Explicit Vectorization

- Currently working on support for nestedSDFG within Maps / ConditionalBlocks
- Replaces all scalar-operations with vectorized call to intrinsics. Copies data to storage required by the vector unit.



Publication Plan For ISCA 2026

- We see ISCA as our first goal (Deadline 17th of November). If we cannot complete everything before the deadline, ASPLOS would be a strong alternative (Deadline March 26). If we do not encounter any unexpected blockers, I believe we can target the ISCA deadline.

Publication Plan For ISCA 2026

- Mid-October 2025: Verification pipeline improvements and automation completed.
- End of October 2025: CloudSC integration completed (aligned with the end-of-year-two SoftHier deliverable). This is mainly my responsibility.
- End of October 2025: Re-collection of GEMM and CloudSC data with additional hardware configurations completed. This is mainly Aofeng's responsibility.
- Mid-November 2025: Collection of CloudSC data with additional hardware configurations completed, and paper writing.